Analyzing Fuel Price Trends in USA: Implications for Airline and Railway Industries

Jal Dani (202201315)
Time Series Analysis (SC475), DAU
Gandhinagar, India
202201315@daiict.ac.in

Sumit Vishwakarma (202201320)
Time Series Analysis (SC475), DAU
Gandhinagar, India
202201320@daiict.ac.in

***Abstract*—In this study, we begin by examining the original monthly datasets of US diesel and kerosene prices along with US railways and airlines ticket prices ranging from year 2000 to 2020. Subsequently, we visualize trends, seasonality, and residual components through time series decomposition and further conduct statistical analysis, including mean-variance evaluation, auto-covariance analysis, and stationary tests.**
We then try to identify

- **the relationship between diesel prices ($/gal) and railway ticket prices**
- **the relationship between kerosene prices ($/gal) and flight ticket prices**

by computing cross-covariance and cross-correlation between respective ticket and fuel prices to assess their dependence over the time.

Our ultimate goal is to derive the interdependence between fuel and ticket prices, which helps us better understand the impact of fuel cost fluctuations on both the airline and rail industries. It can also be used in statistical models for time series forecasting, such as ARIMA and SARIMA (seasonal ARIMA), after removing trends and seasonality, if present, to predict future ticket pricing trends based on fuel price variations.

I. Introduction

Airline and Railway are vast sectors in the United States, with their combined operational costs exceeding $70 billion, significantly influenced by fuel expenses, labor costs and aircraft/locomotive maintenance. Among these, fuel costs account for nearly 20 30% of the total expenses, making fuel price fluctuations a critical factor in the profitability of these industries. Using time-series techniques such as cross-correlation analysis, we seek to understand how closely ticket fares follow volatility of fuel prices and how it influences pricing decisions in these industries.

II. Proposed Approach

A. Data Extraction

We processed and corrected the irregular/NaN values present in the monthly datasets of kerosene and diesel prices along with the fares of airline and railway tickets.

B. Data Plotting

We plotted the data points of each data set with the y-axis values as the price in USD and the x-axis as the time range from 2000 to 2020.

C. Time Series Decomposition

Considering the best model between additive and multiplicative, we performed the time series decomposition on each data set to plot and visualize the trend, seasonal and residual components.

1) Trend Identification: The trend component shows the long-term direction in the data. In our plot, this component highlights the impact of market forces and other economic factors such as recession on the short-term fluctuations of fuel and travel prices.

2) Seasonal Component: This component helps to depict any repeating patterns or variations in our datasets, we plotted the seasonal component.

3) Residual Component: This component is left over after the removal of the trend and seasonal components. Indicates any outliers or sudden and unpredictable fluctuations.

D. Stationary Check

Most forecasting methods like ARIMA works on stationary data. In order to check Stationary, first we plotted the mean and variance of our dataset to visually identify the constant mean and variance. Additionally, we applied AD-Fuller (ADF) test, to confirm non-stationary in our data.

E. De-trending

In order to better analyze the data, we de-trended the time series using polynomial regression method to then focus on the remaining residual and seasonal components.

F. De-seasonalizing

Upon detrending the series, we removed the seasonal component to further focus on the residual and study the causes of unpredictable and sudden changes in the time series.

G. Confirm Stationarity

After performing the time series decomposition, we confirmed the stationarity of process by observing the constant mean and variance plot.

H. Cross Correlation Analysis

In order to depict the possible dependence of fuel prices on travel ticket fares, we analyzed their cross-correlation plot with a lag of 24 months.

III. Plots and Implications

A. U.S. Diesel Prices - Trend, Seasonality and Residual Plots

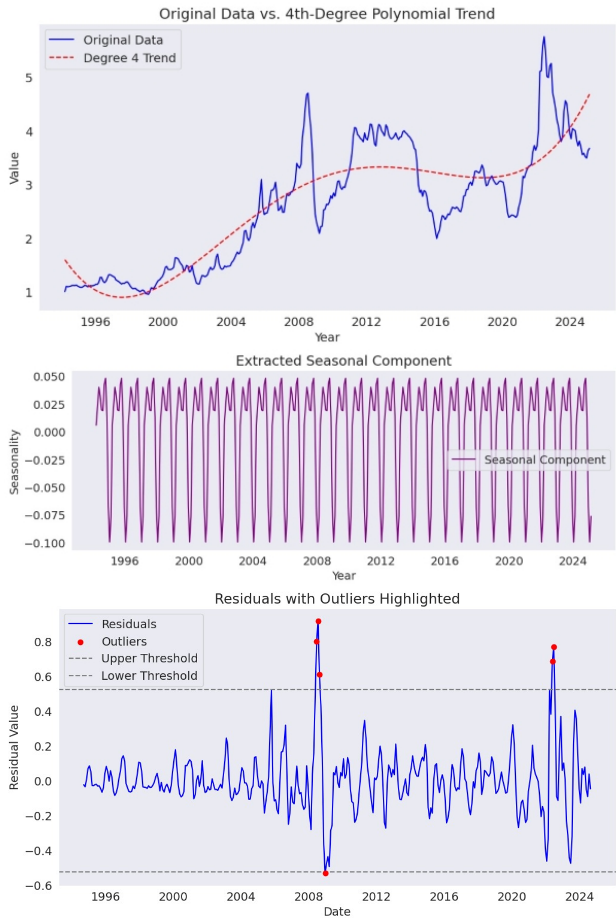

Fig. 1. Plots for US Diesel prices from 1996 to 2024.

a. Decomposition analysis:

- **Trend:** From the plot, it is evident that the trend shows a long-term increasing pattern with notable spikes and dips at certain points, whose reasons are discussed as the next section.
- **Seasonality:** A repeating pattern can be observed from the seasonal component plot which may be possibly influenced by Oil market behavior.
- **Outliers in the Residual** The residual plot has outliers (data points outside of the confidence interval of -3 to +3 sigma region), at two time periods, particularly from 2007 to 2009 and from 2022 to 2023, which indicate unpredictable events affecting the time series at that particular point of time.

b. Abnormalities and Their Possible Causes:

- **Great Recession - 2007 to 2009 :** The Diesel prices were significantly increased in 2007 before entering the period of recession. As the economic uncertainties increased in 2008, it resulted in drastic fall of prices due to the minimal demand resulting from unemployment and nominal business spending across the nation.
- **Surge in U.S. shale oil production - 2014 :** From the June of 2014, the kerosene and other oil prices across the globe hit an all time highest drop since the World War 2. Main reasons for this includes the surge in U.S. shale oil production and some key policy changes by the Organization of Petroleum Exporting Countries (OPEC).
- **Russia - Ukraine War 2022 :** The Diesel prices saw significant fluctuations from 2022 due to the escalation of conflicts and tensions between the two countries, resulting in economic sanctions and supply uncertainties.

B. U.S. Railway Fares - Trend, Seasonality and Residual Plots

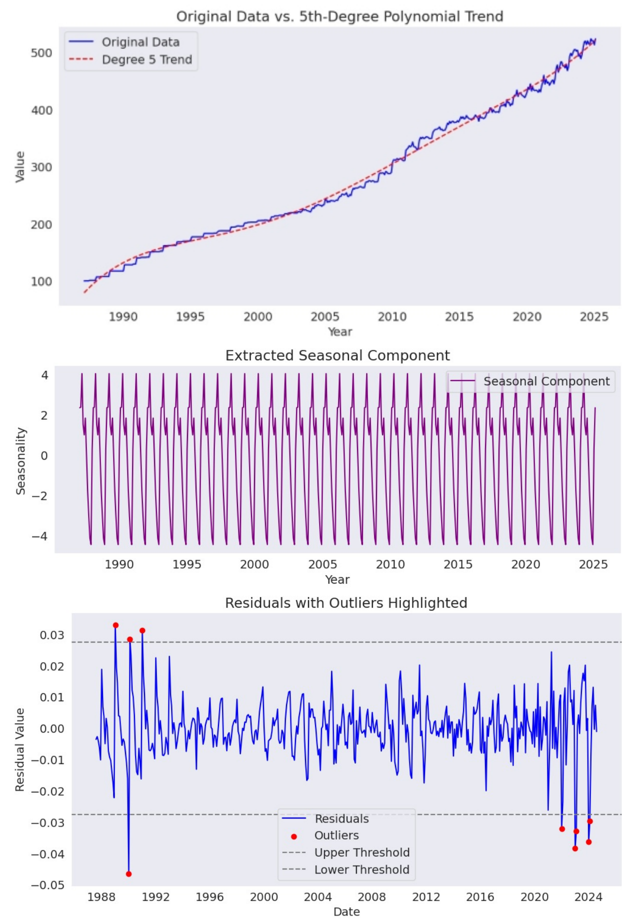

Fig. 2. Plots for US Railway fares from 1980 to 2024.

a. Decomposition Analysis:

- **Trend:** From the plot, it is evident that the trend shows a long-term increasing pattern with no clear spike or dip of any kind at any point.
- **Seasonality:** A repeating pattern can be observed from the seasonal component plot which may be possibly due to seasonal ticket demand based on passengers' tourism pattern.

- **Residual and Outliers:** The residual plot has outliers at two time periods, first from 1990 to 1992 and the next from 2022 to 2024.

b. Abnormalities and Their Possible Causes:

- **COVID-19 Relaxed Restrictions:** From 2022, the restrictions which were imposed on traveling, were significantly reduced which resulted in spike in number of passengers and hence, the fares.

IV. Mean, Variance and ACF Plots for Stationary Check

A. U.S. Diesel Prices - Mean, Variance and ACF Plots

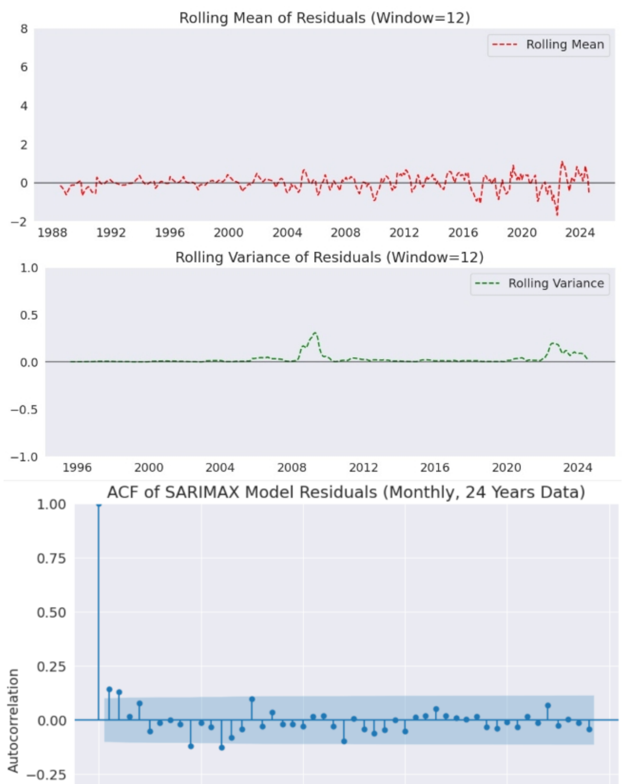

Fig. 3. Plots for US Diesel prices from 1996 to 2024.

a. Decomposition Analysis:

- **Mean:** Upon decomposition, the mean of the residuals remain close to zero, which indicates that there is no continuous increasing or decreasing pattern.
- **Variance:** The variance is also found to be constant at almost entire time period which suggests minimal fluctuations in diesel prices around the mean.
- **Auto-Correlation function:** After carefully observing the plot, it can be said that there is no strong fluctuation or pattern left in the residuals as mostly all the points lie inside the confidence interval.

B. U.S. Railway Fares - Mean, Variance and ACF Plots

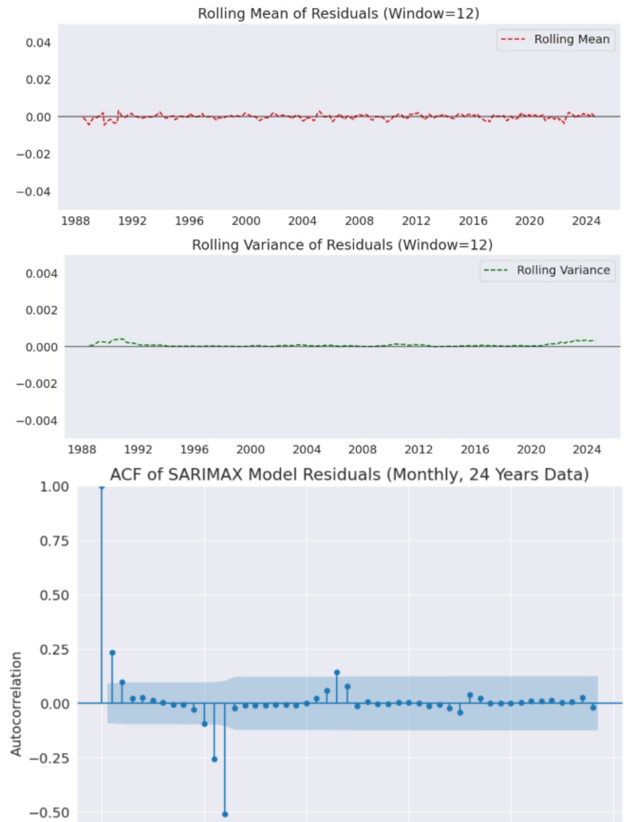

Fig. 4. Plots for U.S. Railway Fares prices from 1980 to 2024.

a. Decomposition Analysis:

- **Mean:** Upon decomposition, the mean of the log-residual remain close to zero, which indicates that there is no continuous increasing or decreasing pattern.
- **Variance:** We plot the variance of log-residual due to higher degree of fluctuations of variance in the residual upon decomposition. The variance of log residual seems stable across entire time period.
- **Auto-Correlation function:** After carefully observing the plot, there are few time periods where there is strong fluctuation. Still, mostly all the points lie inside the confidence interval.

V. Quantile Analysis

A. U.S. Diesel Prices

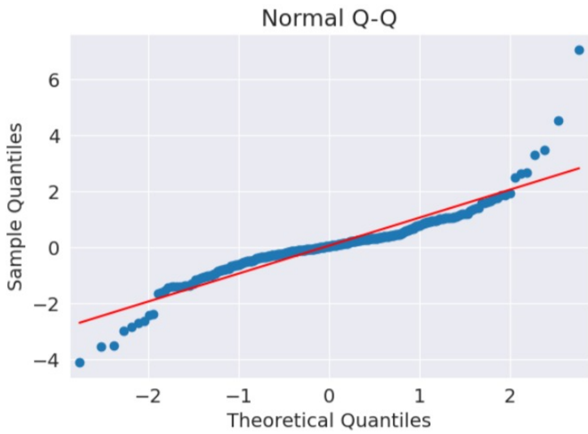

Fig. 5. Q-Q Plot for U.S. Diesel prices

Plot Analysis:

- **Left Tail:** The left tail is observed to be lower due to great recession from 2007 to 2009 and Global Oil crisis of 2014.
- **Right Tail:** The right tail is observed to be higher due to Russia - Ukraine war in 2022, which resulted in significant increase of diesel prices in U.S.
- **Extreme Outliers:** We observe outliers at both the ends, which is due to strong fluctuations in the initial time period (2007-2009) and latest time period (2022-2024).

B. U.S. Railway Fares

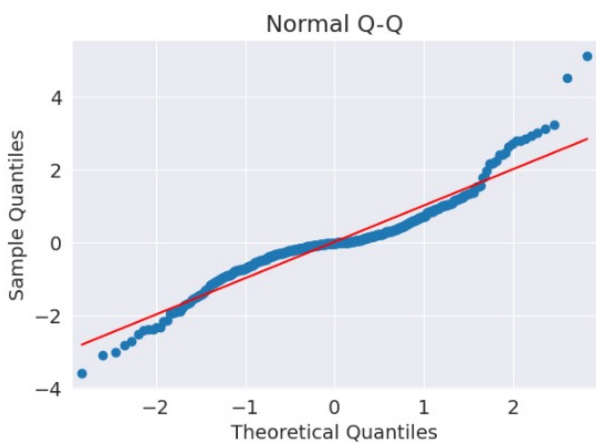

Fig. 6. Q-Q Plot for U.S. Railway Fares

Plot Analysis:

- **Left Tail:** The left tail is observed to be lower due to great recession from 2007 to 2009 and Global Oil crisis of 2014.
- **Right Tail:** The right tail is observed to be higher due to Russia - Ukraine war in 2022, which resulted in significant increase of diesel prices in U.S.
- **Extreme Outliers:** We observe outliers at both the ends, which is due to strong fluctuations in the initial time period (2007-2009) and latest time period (2022-2024).

VI. Distribution Fitting

A. U.S. Diesel Prices

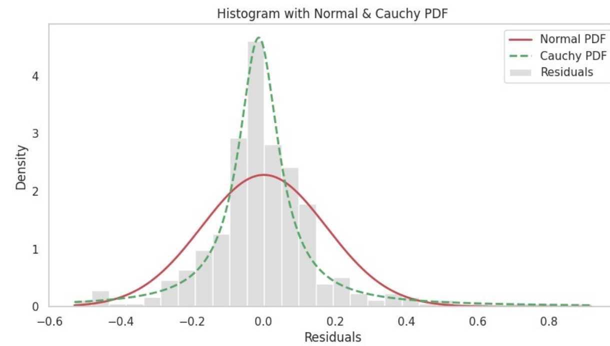

Fig. 7. Distribution Fitting for U.S. Diesel prices

Plot Analysis:

- **Diesel Plot:** The Residuals deviate from the Normal Distribution. We conclude that it nearly follows a Cauchy Distribution and hints some irregular and unpredictable fluctuations.

B. U.S. Railway Fares

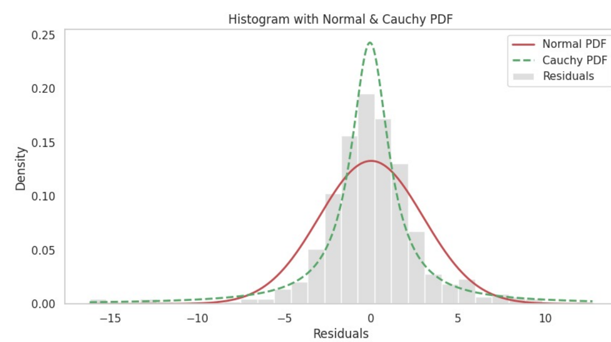

Fig. 8. Distribution Fitting for U.S. Diesel prices

Plot Analysis:

- **Railway Plot:** The Residuals deviate from the Normal Distribution. We conclude that it nearly follows a Cauchy Distribution and hints some irregular and unpredictable fluctuations.

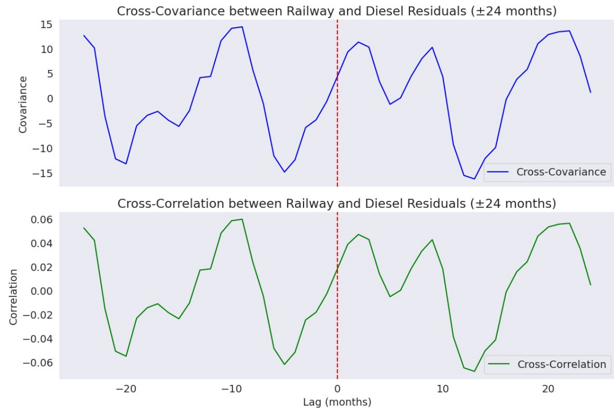

Fig. 9. Cross Covariance and Correlation Plot - 24 month gap

VII. Cross Covariance Analysis

A. 24 months gap

Cross Covariance Plot Analysis:

- **24 month window:** We try to find any cyclical or repeating patterns in the CCF through analyzing a 24 month window.
  As observed, the multiple peaks and dips may suggest that the interdependence of the prices of Diesel and Railway tickets may be seasonal.
  A positive correlation at lag of +1 to +8 months implies that the cost of railway tickets increased shortly after an increase in diesel prices, which is technically correct as over 20% to 30% of operational costs in Railway industry depends on fuel prices.
  This concludes that these both are not completely independent and have a short term impact on each other.

B. COVID-19 Time Period

Plot Analysis:

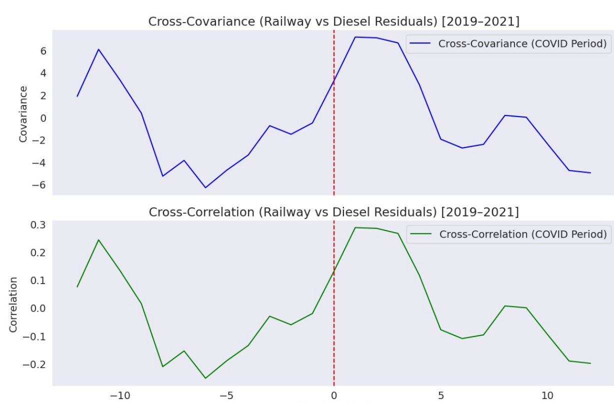

Fig. 10. Cross Covariance and Correlation in COVID-19 Period

- **COVID-19 Period:** This plot focuses on the COVID-19 time period and try to deduce if the prices of Diesel impacted the Railway fares. If yes, then how and till which time (lag period).
  Upon studying the plot, we observe moderate positive correlation at lag +1 to +3 months, suggesting diesel price changes may have influenced railway activity shortly afterwards any increase or decrease in Diesel Prices.
  This Short window reflects a more uncommon relationship between them which may likely be caused by pandemic disruptions and restrictions.

VIII. Conclusion

From this study, we analyzed how fluctuations in fuel prices (diesel and kerosene) and travel fares in USA depend on each other, using time series decomposition and cross-correlation techniques. We identified clear trends and seasonal patterns, along with notable outliers linked to major economic events like The Great Recession of 2008 and COVID-19 outbreak period of 2020 to 2022. Our findings indicate that changes in diesel prices are followed by short-term adjustments in railway ticket prices, highlighting a dependent relationship. Overall, this work lays the groundwork for improved forecasting models in the transportation industry.

Resources

All the codes used in this study, along with additional analysis of kerosene and flight data, are available at the following Google Drive link:

- **Project Code and Data:** The code used in this analysis, along with additional data, is available at:
  Google Drive - Project Code (Pynb) and Resources

References

[1] U.S. Energy Information Administration, "Gasoline and Diesel Fuel Update," [Online]. Available: https://www.eia.gov/petroleum/gasdiesel/.

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[3] U.S. Energy Information Administration, "Petroleum & Other Liquids – Kerosene-Type Jet Fuel Prices," [Online] Available: https://www.eia.gov/dnav/pet/hist/LeafHandler.ashx?n=PET&s=EMA_EPPK_PWG_NUS_DPG&f=M